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6440-BS-SE-23
Assignment#2

Q:Find out the page size of different Os?

Windows :

In Microsoft Windows operating systems, memory is managed in units called pages. The default **page size is 4 KB on x86 and x64 architectures**. Windows also supports larger page sizes to improve performance for memory-intensive applications. For instance, with Physical Address Extension (PAE) enabled, x86 systems can use 2 MB pages, while x64 systems support 2 MB pages by default. On IA-64 architectures, the small page size is 8 KB, and large pages can be 16 MB. These configurations help reduce Translation Lookaside Buffer (TLB) misses and enhance overall system performance. However, using large pages can lead to increased internal fragmentation, so they are typically reserved for specific applications that can benefit from them. For more detailed information, refer to the Windows Internals documentation:

Source:WindOh.com - Paragraph#13,Line#2

URL: https://windoh.wordpress.com/2010/01/04/the-windows-memory-architecture-part-1-the-basics/?utm_source=chatgpt.com

Linux:

In Linux operating systems, memory management is structured around pages, **with the default page size typically set at 4 KB** across most architectures. This standard size is favored for its efficient balance between memory utilization and system performance. For applications requiring larger memory blocks, Linux supports huge pages, which can be 2 MB or 1 GB in size, depending on the CPU architecture and configuration. Utilizing huge pages can reduce the overhead associated with managing numerous small pages, thereby enhancing performance for memory-intensive applications. The availability and configuration of huge pages are contingent on both hardware support and kernel settings. For detailed information on huge pages and their implementation in Linux, refer to the Linux Kernel documentation: .

Source:lambdafunc.medium.com- (Paragraph#3, Line#1)

URL: <https://lambdafunc.medium.com/linux-memory-management-2c13b5e386f1>

Android:

In Android operating systems, memory management traditionally utilizes a **4 KB page size**, aligning with most CPU architectures. However, with the introduction of Android 15, Google has incorporated support for a larger **16 KB page size**. This enhancement aims to improve system performance by reducing the frequency of memory address translations, thereby decreasing Translation Lookaside Buffer (TLB) misses. The adoption of a 16 KB page size can lead to performance gains of approximately 5–10%, albeit with a slight increase in memory usage. To facilitate this transition, Android 15 includes developer options and emulator support for testing applications with the new page size. Starting November 1, 2025, Google Play will require all new apps and updates targeting Android 15 or higher to support 16 KB page sizes, ensuring compatibility with evolving hardware standards.

Source:Android Developers Blog.com – (paragraph#1, Line#3)

URL: https://android-developers.googleblog.com/2024/12/get-your-apps-ready-for-16-kb-page-size-devices.html?utm_source=chatgpt.com

MacOS

In macOS, the system's memory management employs a paging mechanism where memory is divided into fixed-size blocks called pages. The default page size is **typically 4 KB** (4,096 bytes), aligning with the standard page size used by many Unix-like operating systems. This page size is determined by the underlying hardware architecture and is not generally configurable by the user at runtime. For applications requiring larger memory blocks, macOS supports the use of larger pages, such as **2 MB or 1 GB**, through mechanisms like huge pages, which can enhance performance for memory-intensive tasks. The `getpagesize()` function in macOS returns the system's page size, providing a programmatic way to retrieve this information.

Source:Developer.Apple.com – (paragraph#1, Line#3)

URL:https://developer.apple.com/library/archive/documentation/System/Conceptual/ManPage_s_iPhoneOS/man5/manpages.5.html#/apple_ref/doc/man/5/manpages

Chrome:

Google Chrome, as a web browser, operates on various operating systems such as Windows, macOS, and Linux. The memory management and page size utilized by Chrome depend on the underlying operating system and hardware architecture. For instance, on 64-bit systems, the default page size is **typically 4 KB**, aligning with the standard page size used by many operating systems. Chrome's memory management involves dividing virtual memory into pages, with each page representing a fixed-size block of memory. This allows for efficient allocation and

management of memory resources, enabling Chrome to handle multiple tabs and processes concurrently. However, specific details about Chrome's memory management and page size may vary depending on the operating system and hardware configuration.

Source: chromium.googlesource.com - (Paragraph#2, line 1)

URL:https://chromium.googlesource.com/chromium/src/%2B/lkgr/docs/memory/key_concepts.md?

UNIX:

In UNIX and UNIX-like operating systems, the default memory page size is **typically 4 KiB** (4,096 bytes), which is determined by the architecture of the CPU. This standard page size is commonly used across various systems, including Linux. For example, on x86 systems, the basic page size is 4 KiB, and this is a property of the CPU architecture. On 64-bit ARM systems, the page size can be configured at build-time as **4 KiB, 16 KiB, or 64 KiB**, depending on the specific requirements and configuration of the system .

Source: unix.stackexchange.com-Page#1

URL: <https://unix.stackexchange.com/questions/495987/how-does-the-operating-system-read-the-page-size-from-hardware?>
